PROPER LIMIT OF PROPER FINITE PRESENTATION MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Chow's lemma. Cohomology of Schemes, Chow lemmas
- Proper image. Morphisms, proper image lemmas

Source reference. Original proposition 01ZJ: sources/stacks-chow/tags.txt, source lines 2596–2596. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0202: sources/stacks-chow/tags.txt, source lines 2619–2619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZP: sources/stacks-chow/tags.txt, source lines 12607–12607. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0A0N: sources/stacks-chow/tags.txt, source lines 12641–12641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a proper morphism with S quasi-compact and quasi-separated. Then $X = \lim X_i$ is a directed limit of schemes X_i proper and of finite presentation over S such that all transition morphisms and the morphisms $X \to X_i$ are closed immersions.*

Proof. By Proposition 01ZJ we can find a closed immersion $X \to Y$ with Y separated and of finite presentation over S. By Lemma 0202 we can find a diagram

$$\begin{array}{ccccc} Y & \xleftarrow{\pi} & Y' & \longrightarrow & \mathbf{P}^n_S \\ & \searrow & \downarrow & \swarrow & \\ & & S & & \end{array}$$

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where $Y' \to \mathbf{P}^n_S$ is an immersion, and $\pi : Y' \to Y$ is proper and surjective. By Lemma 09ZP we can write $X = \lim X_i$ with $X_i \to Y$ a closed immersion of finite presentation. Denote $X'_i \subset Y'$, resp. $X' \subset Y'$ the scheme theoretic inverse image of $X_i \subset Y$, resp. $X \subset Y$. Then $\lim X'_i = X'$. Since $X' \to S$ is proper (Morphisms, Lemmas 01W3), we see that $X' \to \mathbf{P}^n_S$ is a closed immersion (Morphisms, Lemma 01W6). Hence for i large enough we find that $X'_i \to \mathbf{P}^n_S$ is a closed immersion by Lemma 0A0N. Thus X'_i is proper over S. For such i the morphism $X_i \to S$ is proper by Morphisms, Lemma 03GN. $\square$